Australian International Trade, 2016 to 2025

What ten years of trade data say about Australian fuel security

32146 Visual Analytics, Assessment Task 3. Student 24503993. Dashboard and storyboard: https://REPLACE-ME.github.io/datavis-a3/

1. Executive summary

Australian export grew from $258,469m in 2016 to $523,098m in 2025. Import grew from $254,345m to $460,866m. The trade balance looks healthy at first sight, but it peaked at $176,863m in 2022 and then fell 65% to $62,232m by 2025. The surplus of the last decade rested on a single price event and it is draining away.

Main category 3, mineral fuels, lubricants and related materials, is the largest mover in seven of the nine year transitions in the dataset. It sets the direction of the decade in both directions of trade. Inside it sits the finding this report is built around.

Subcategory 33, petroleum, is the only large part of Australian trade where the country buys far more than it sells. Export grew 104% across the decade and import grew 126%, so the deficit widened from $16,369m to $38,555m. Import and export move together at a correlation of 0.91, because both are priced off the same world oil market. Import is about three times the size of export, so every oil price rise widens the deficit rather than closing it. Australia cannot export its way out of this gap.

The recommendation is a target, not a forecast. Cutting petroleum import 3% a year from 2025 reaches $45,291m by 2030. The straight line fit on ten years of data has an R squared of only 0.59, so no five year point forecast on this series deserves confidence. Section 12 shows an AI model, given the same ten numbers, forecasting the opposite direction.

2. Data collection and preparation

The data is the Australian International Trade dataset from the Australian Bureau of Statistics, supplied as 2026-Spring-Ass3-V20.xlsx. It holds two sheets, one for export and one for import, covering 2016 to 2025. Each sheet is wide: row 1 names the category, row 2 gives the unit, row 3 gives the ABS series identifier, and rows 4 to 13 hold the ten years. There are 10 main categories, 67 subcategories and one total column, which is 78 series in each direction.

The source file was never edited. A Python script reads it and writes a new workbook, 24503993_A3.xlsx. Four checks run on every build: the extract must produce 1,560 rows, all ten years must be present in both directions, every unit must read $ Millions, and the ten main categories must sum to the published total. That last check passes within 6 million dollars in the worst year, which is 0.0013% of the total and is consistent with ten independently rounded series. There are no blank cells anywhere in the source.

Every analytical and statistical result in this report lives in the workbook as a live Excel formula, not as a pasted number. A verification script recomputes growth, index, share, CAGR, the descriptive statistics, the trend fits, the z-scores, the correlation matrices, the change points and the forecasts independently, then compares them against the recalculated spreadsheet to six decimal places. The build fails if any cell disagrees or holds an error value.

3. Dataset characteristics and data types

The dataset mixes several data types and each one drives a different visual choice.

- Year is temporal and ordered. It drives a real axis on every time series in this report.
- Main category and subcategory are nominal, in a two level hierarchy. SITC codes with one digit are main categories and codes with two digits are their subcategories. The hierarchy is what sections 6 and 7 analyse.
- Direction is binary, import or export. Every criterion in this task asks for both, so direction is encoded by colour on every chart and never left implied.
- Value is continuous, in millions of Australian dollars. It is a value, not a volume, which is the single most important limit on what this dataset can prove.

- Derived measures add more types: growth and share are ratios, the index is a rebased ratio, the z-score is a standardised score and the correlation is a bounded coefficient.

The value limit deserves emphasis. The dataset records dollars, so it cannot separate a price change from a volume change. Where this report states a cause it says whether the evidence supports a price reading or a volume reading, and it does not claim a volume effect that the data cannot show.

4. The patterns built in Excel, and how they reach the dashboard

The workbook holds 23 sheets and 19 native Excel time series charts. Each chart carries a title and a label on both axes. The analysis sheets are these.

Sheet	Pattern it holds	Type
Main-Analytics	Growth, index, share, CAGR, contribution to growth	Analytical
Main-Stats	Mean, median, spread, coefficient of variation, trend fit, z-score	Statistical
Balance	Export minus import, and the cover ratio	Analytical
Correlation	Two 10 by 10 correlation matrices on growth	Statistical
ChangePoints	Largest rise and fall in each year, with a cause	Analytical
Cat3-SubAnalytics	Main category 3 against its subcategories	Analytical
Pet33-Focus	The focus subcategory, with an insight for each year	Analytical
Pet33-vs-Sub	Petroleum against coal and gas, with correlations	Statistical
Pet33-Forecast	Linear trend and CAGR forecasts, and the target path	Statistical
AI-Comparison	My forecast against the AI forecast	Analytical

Deployment into the application is literal rather than claimed. The dashboard and the storyboard do not recompute anything. A reader module opens the recalculated workbook and reads the finished values out of these sheets, so a marker who changes a cell in the spreadsheet and rebuilds will see the charts move. The spreadsheet is the single source of every number in this report and on both web pages.

One implementation note is worth recording. The openpyxl library writes modern Excel function names without the _xlfn prefix that the file format requires, so LibreOffice returns #NAME? for them. Every formula in the workbook therefore uses a legacy function name, such as FORECAST and STDEVP rather than FORECAST.LINEAR and STDEV.P. Both spellings are valid in current Excel, and the legacy names work in both applications.

5. All ten main categories: raw, analytical and statistical trends

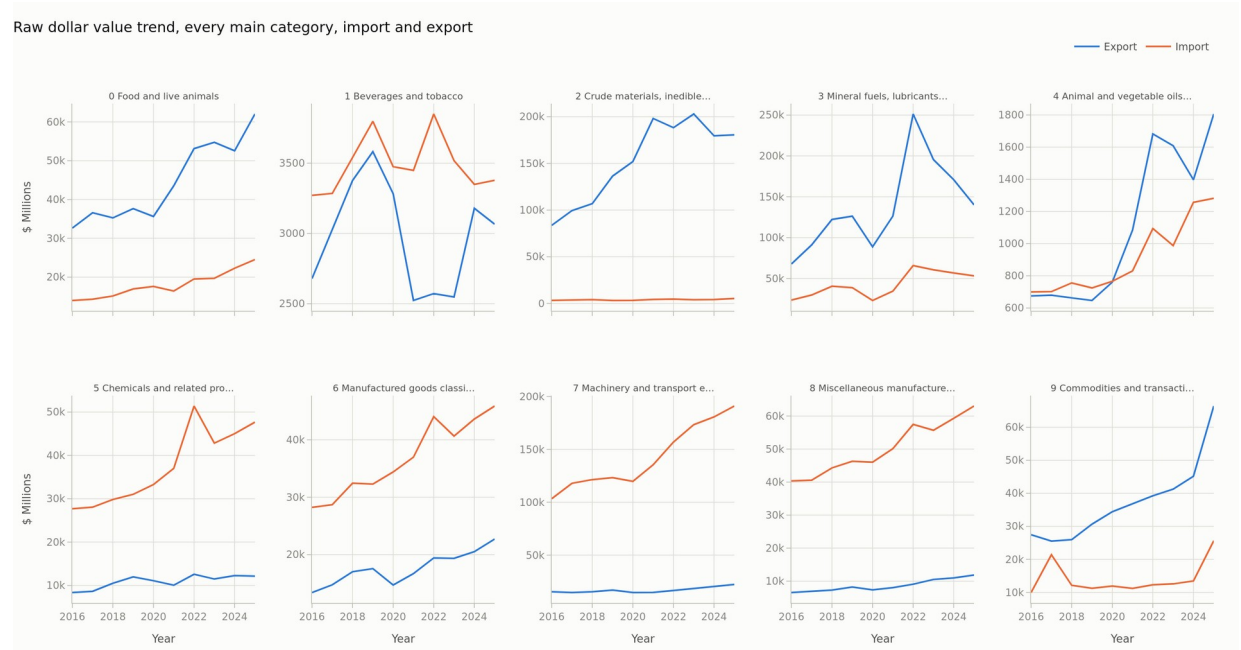


Figure 1. Raw dollar value trend for every main category, both directions, in $ Millions. One panel per category so that small categories stay readable next to large ones.

Ten categories cannot share one set of colours without becoming unreadable, so the raw view uses one panel per category and keeps colour for the one distinction that matters everywhere in this report, which is import against export. The export side is concentrated. Category 2 contributed $97,030m of export growth and category 3 contributed $72,479m. The import side is broader, and category 7 alone contributed $87,624m.

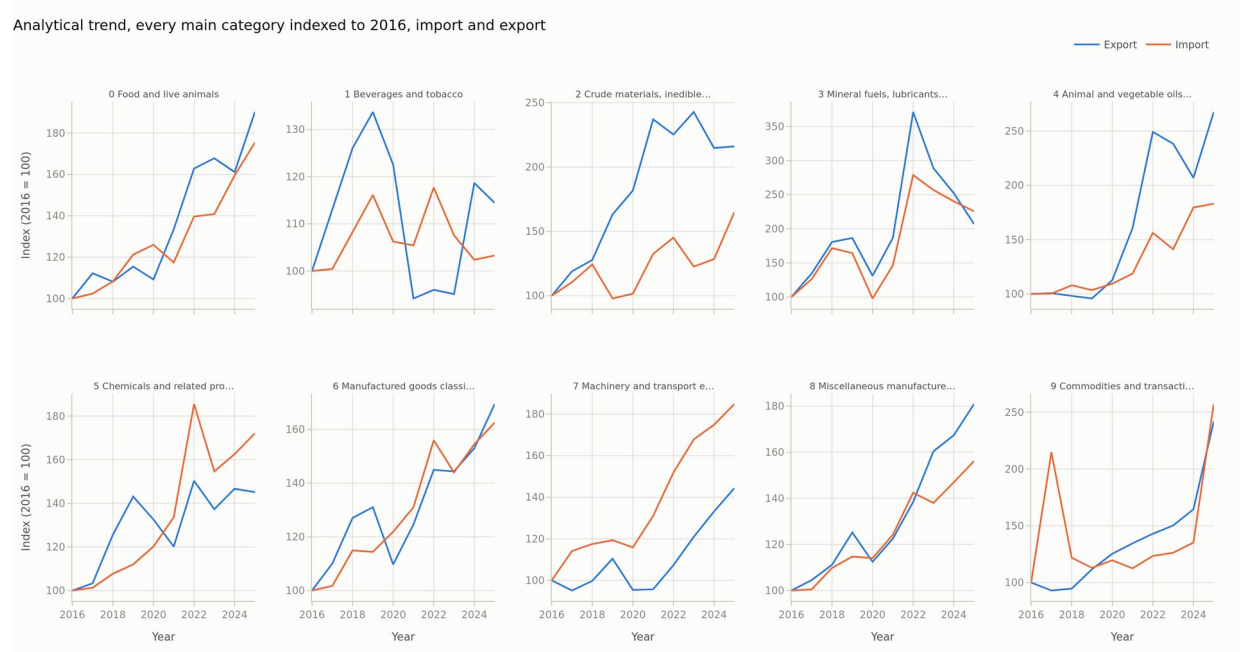


Figure 2. Analytical trend. The same ten categories with 2016 set to 100, which removes size and compares growth rate instead.

Indexing reorders the winners. Category 4 is the smallest category in dollars but grew fastest, at a CAGR of 11.6% on the export side. Category 1 is the laggard at 1.5%, and it is the only category whose export ends the decade near where it started. A raw chart hides both facts behind the scale of categories 2 and 3.

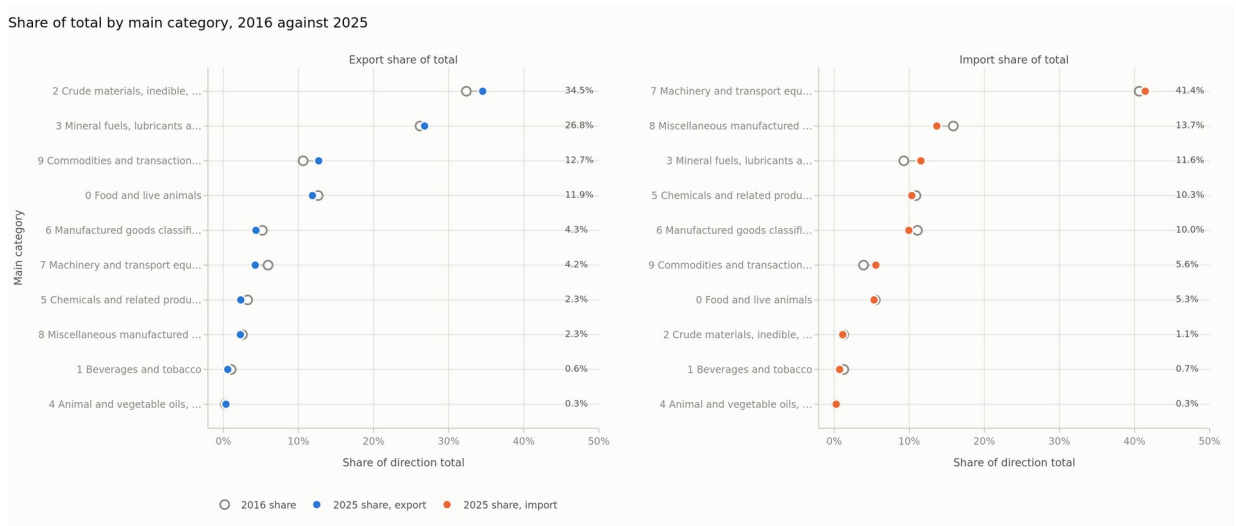


Figure 3. Each main category as a share of its own direction total, in 2016 and again in 2025.

The export mix shifted and the import mix did not. Category 2 rose to 34.5% of all export by 2025. On the import side category 7 held about 41% across the whole decade. Australia changed what it sells considerably more than it changed what it buys.

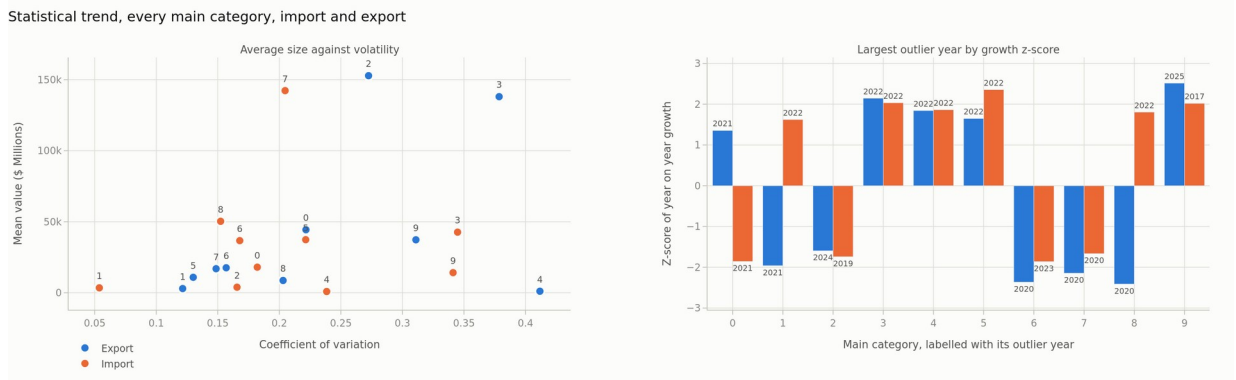


Figure 4. Statistical trend. Average size against the coefficient of variation on the left, and the single most unusual growth year for each category on the right, measured as a z-score.

The left panel separates the steady categories from the volatile ones. Category 3 has the highest export volatility at a coefficient of variation of 0.38, while its trend fit explains only 44% of the variance. A category this volatile cannot be forecast with a straight line, and section 11 returns to that point. The right panel shows that the outlier years cluster: 2020 for the manufacturing categories and 2022 for energy.

6. How the main categories relate, and the change point in each year

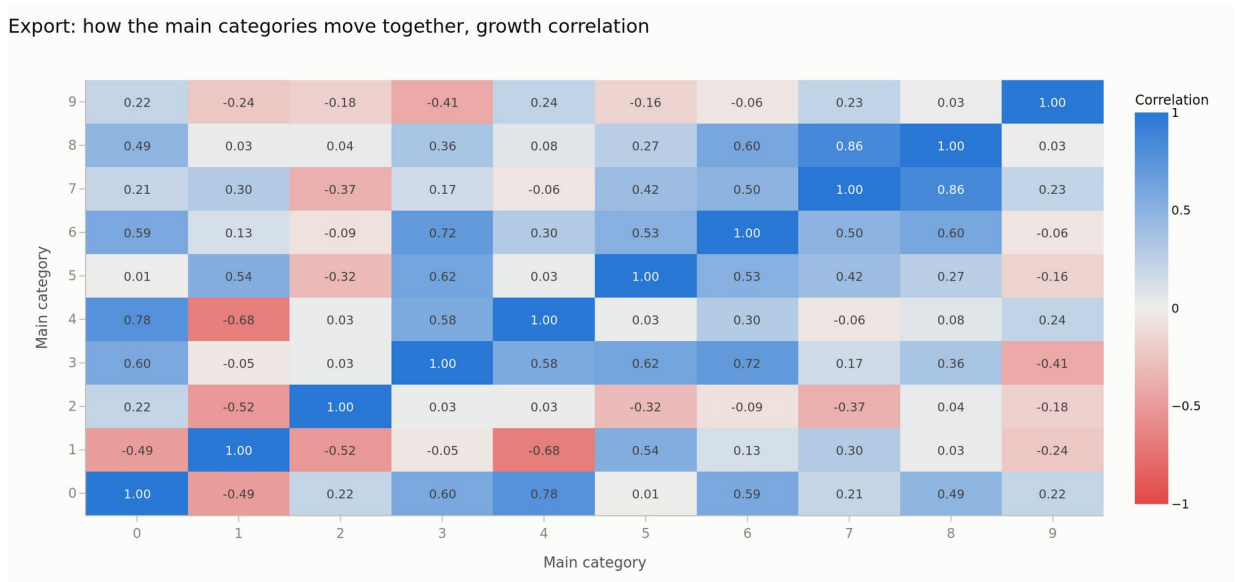


Figure 5. Correlation of year on year growth between every pair of main categories, export side. Blue means the pair moves together and red means it moves apart.

The two largest export categories are unrelated. Category 3 against category 2 is r = 0.03, which is effectively zero across nine growth years. Category 3 instead moves with category 6 at r = 0.72 and with category 5 at r = 0.62, because all three follow the same world price cycle. On the import side category 3 against category 7 is r = 0.76. This matters for policy: the two pillars of Australian export do not hedge each other, because they are driven by separate cycles that happen to overlap.

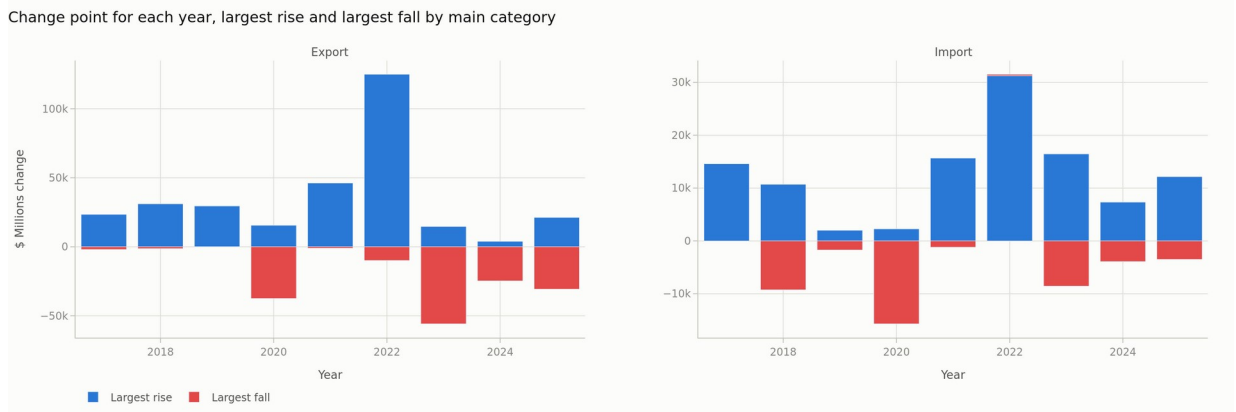


Figure 6. The main category that rose most and the one that fell most in each year, both directions, in $ Millions of change.

Requirement 3 asks for the change point in every year, so the workbook records all nine transitions rather than only the dramatic ones. The table below gives the largest export mover and the cause. Causes drawn from outside the data cite a sourced research note.

Year	Largest export mover	Change	Cause
2017	Mineral fuels	23,447	Mineral fuels export rises 23,447 million on higher coal and LNG prices and new LNG capacity.
2018	Mineral fuels	31,120	Mineral fuels export rises 31,120 million as LNG output reaches full capacity.
2019	Crude materials	29,536	Crude materials export rises 29,536 million.
2020	Mineral fuels	-37,467	COVID cuts mineral fuels export 37,467 million and

			mineral fuels import 15,686 million.
2021	Crude materials	46,211	Crude materials export rises 46,211 million at the iron ore price peak.
2022	Mineral fuels	124,939	Mineral fuels export rises 124,939 million after the invasion of Ukraine.
2023	Mineral fuels	-55,785	Mineral fuels export falls 55,785 million as energy prices retreat from the 2022 peak.
2024	Mineral fuels	-24,707	Mineral fuels export falls 24,707 million and crude materials export falls 23,373 million as energy and iron ore prices both ease.
2025	Mineral fuels	-30,661	Mineral fuels export falls 30,661 million.

Two findings in that table correct a reading that the headline numbers invite. First, China restricted Australian coal from late 2020 until March 2023, yet coal export value rose 124% in 2022, in the middle of the restriction, and fell for three years after it ended. Price, not market access, drove the value, and Australia redirected volume to other buyers. Second, the 2022 fall in iron ore against the rise in coal is not a substitution. The iron ore price had already fallen from its April 2021 record to a low in November 2021, before energy prices spiked in 2022. These are two separate price cycles crossing.

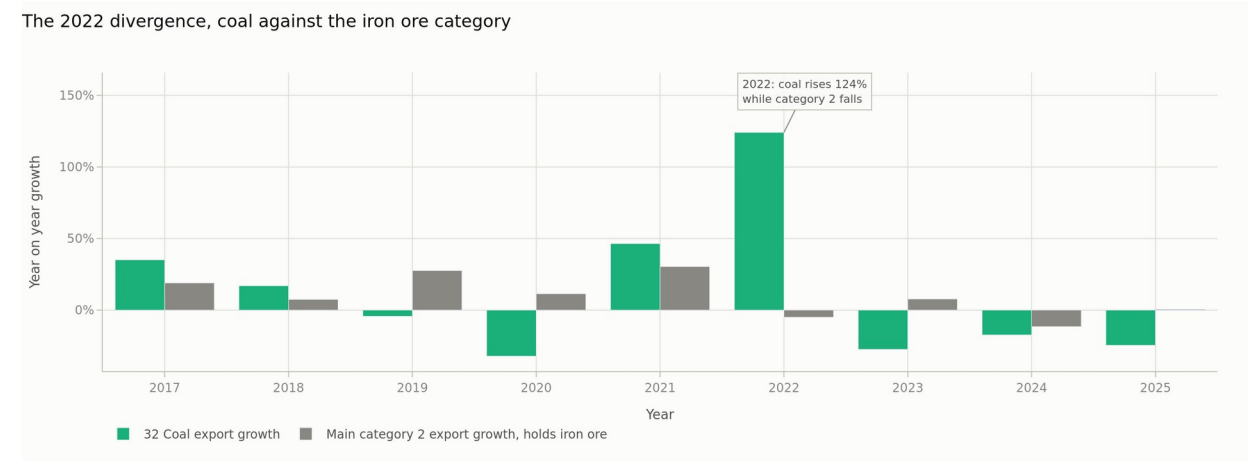


Figure 7. The 2022 divergence. Coal export growth against main category 2, which holds iron ore.

7. Main category against subcategory, inside category 3

The causes in this section are argued from inside the data, as requirement 4 asks. Main category 3 holds exactly three subcategories: 32 coal, 33 petroleum and 34 gas.

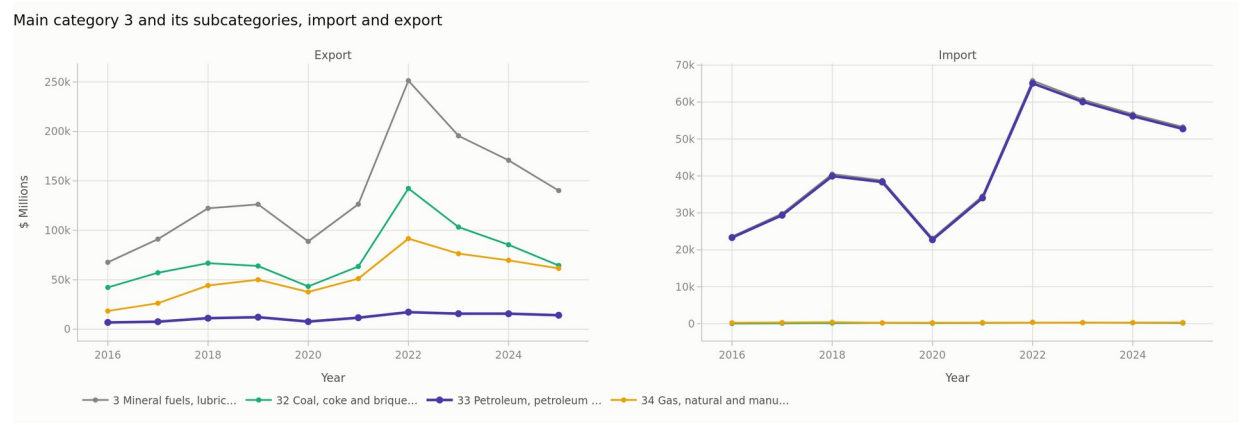


Figure 8. Main category 3 and its three subcategories, both directions, in $ Millions.

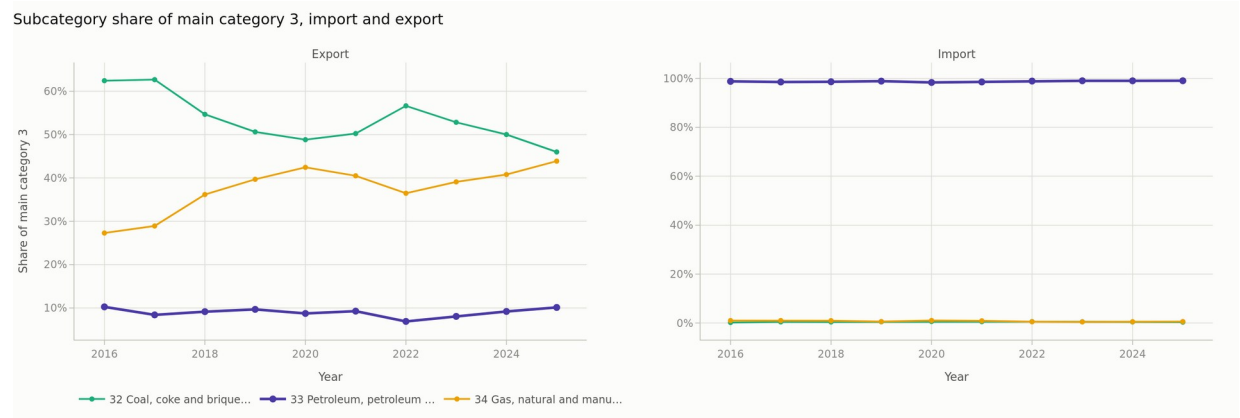


Figure 9. Each subcategory as a share of main category 3, both directions.

The export side and the import side are different categories in practice. On export, petroleum is only 10.1% of category 3 in 2025, behind both coal and gas. On import, petroleum is 99.0% of the category, because Australia imports almost no coal and almost no gas. Any statement about "category 3 imports" is in effect a statement about petroleum alone.

Each subcategory tracks its parent closely on growth: petroleum at r = 0.86, coal at r = 0.99 and gas at r = 0.94. The subcategories do not diversify the main category, they amplify it. The composition did shift, though. Gas rose from 27.3% to 43.9% of category 3 export while coal fell from 62.4% to 46.0%.

One change point can be read from inside the data alone. In 2025 category 3 export fell 18% while petroleum import fell only 6%. If a single demand collapse were driving both sides they would fall together. They do not, so the export fall is a price effect on what Australia sells, and not a fall in what Australia needs to buy.

8. The dashboard

The dashboard is at https://REPLACE-ME.github.io/datavis-a3/dashboard.html and covers subcategory 33 in both directions on one screen. It carries a six tile summary row, five named comparisons, an insight for each of the ten years, and a year range control that re-scales every time chart at once. Each chart has a collapsible table of the numbers beneath it.

- Comparison 1, export against import on one scale, with the 2020 and 2022 change points annotated on the plot rather than left to the caption.
- Comparison 2, the trade balance as columns, every one of them negative.
- Comparison 3, petroleum against its own main category, which exposes the asymmetry between the two directions.
- Comparison 4, petroleum against coal and gas on a common index.
- Comparison 5, import against export as a scatter with the fitted line, which is where the central finding of this report becomes visible.

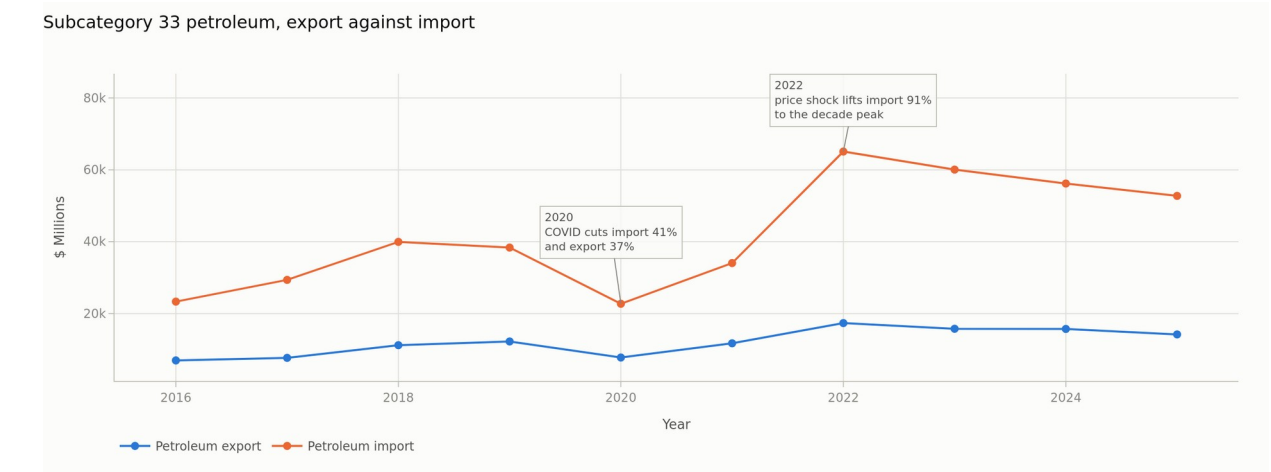


Figure 10. The dashboard anchor chart. Subcategory 33 petroleum, export against import, in $ Millions.

Import exceeds export in every year of the decade. The 2020 dip and the 2022 peak appear on both lines at once, which is the first visual clue that a shared price drives both sides. Import fell 6% in 2025 to $52,742m, which is still 2.3 times the 2016 level.

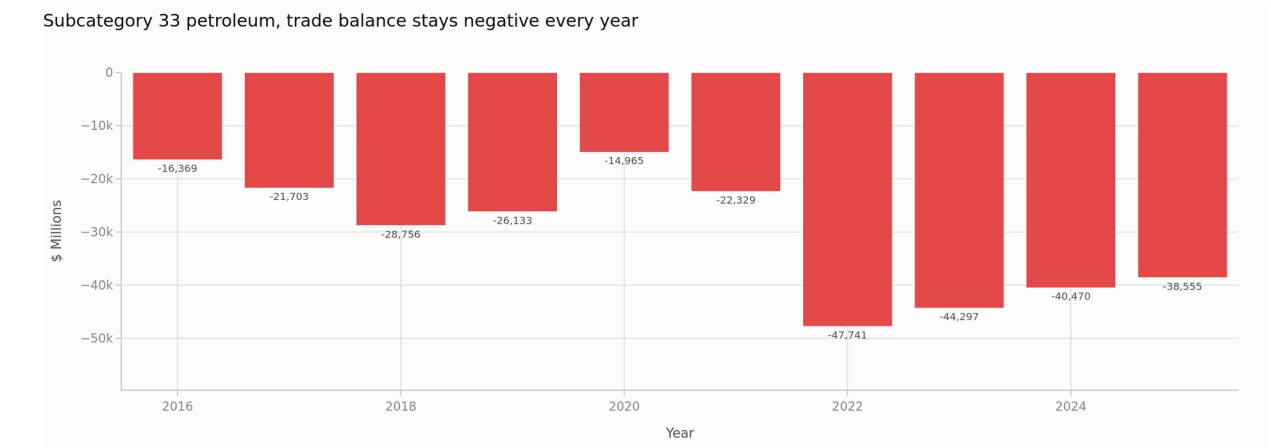


Figure 11. The petroleum trade balance for each year, in $ Millions. Every column points down.

The deficit widened from $16,369m in 2016 to $38,555m in 2025, and reached its worst point of $47,741m in 2022. The cover ratio in 2025 is 0.27, which means each dollar of petroleum import is matched by 27 cents of petroleum export.

9. The storyboard

The storyboard is at https://REPLACE-ME.github.io/datavis-a3/storyboard.html. Where the dashboard answers questions in any order, the storyboard makes one argument in a fixed order across thirteen panels. It starts with total Australian trade, moves through the ten main categories in their raw, indexed and statistical forms, works through the change point of each year, descends into main category 3, then into subcategory 33, and closes with the forecast and the AI comparison. Every panel carries both trade directions and ends with a stated takeaway, so a reader cannot reach the end without having been told what each chart is for.

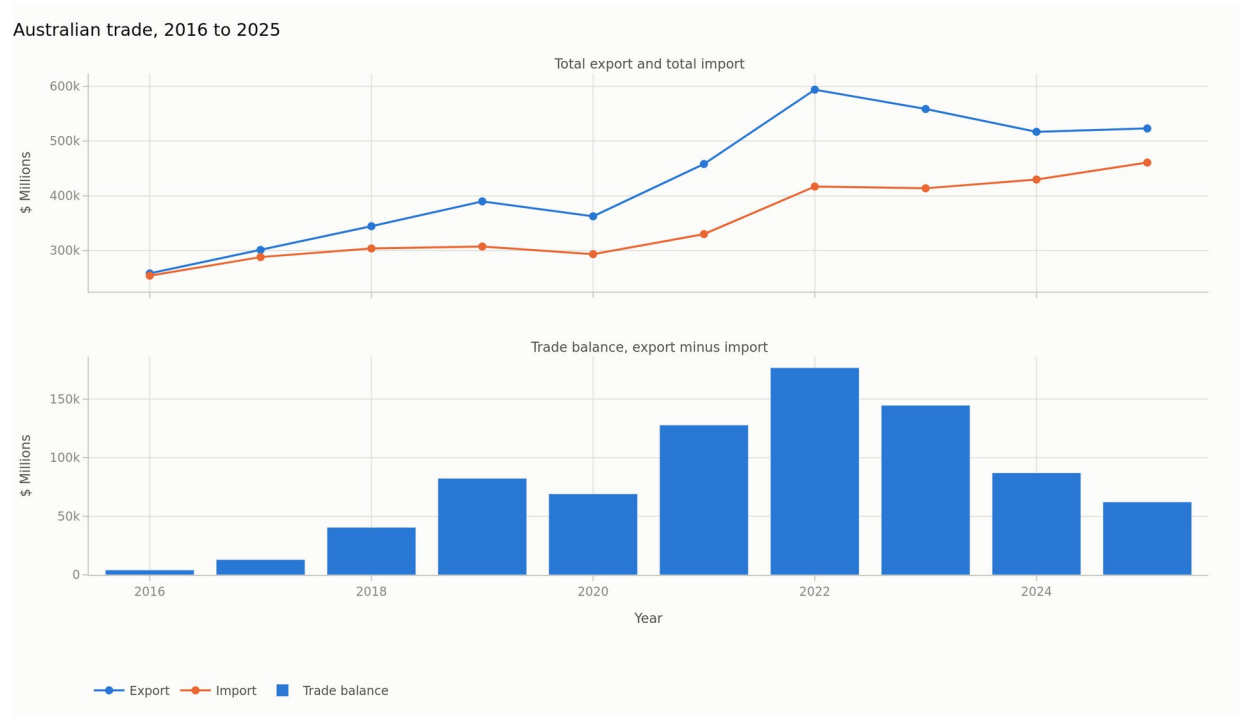


Figure 12. The storyboard opening. Total export, total import and the balance between them, in $ Millions.

This is the panel that sets up everything after it. The balance grew from $4,124m in 2016 to $176,863m in 2022, then fell to $62,232m by 2025. Export retreated from its 2022 peak while import carried on rising without a pause. The surplus is narrowing from both ends.

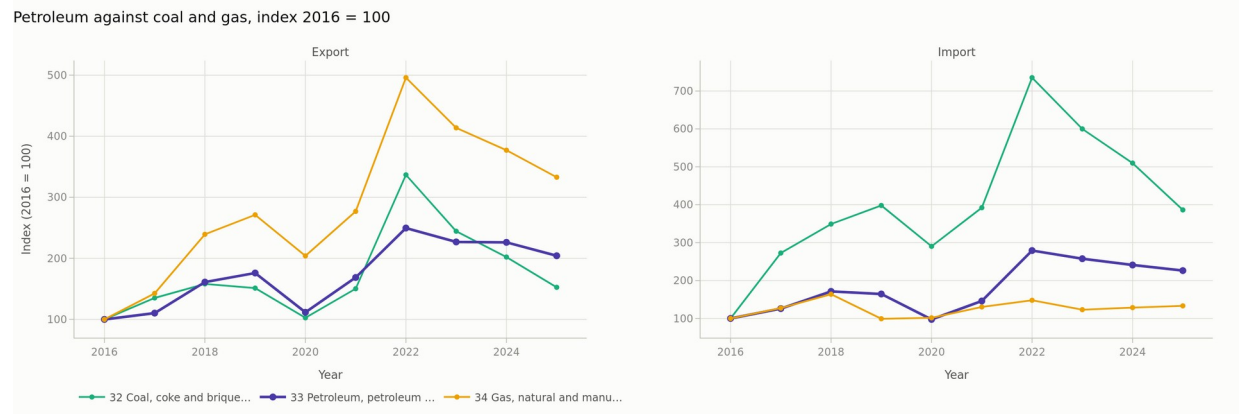


Figure 13. Petroleum against coal and gas, indexed to 2016, both directions.

Petroleum growth tracks coal at r = 0.79 and gas at r = 0.89 on the export side. On the import side those fall to 0.62 and 0.52, because the coal and gas import series are tiny and noisy. The export side of category 3 is one coherent energy story and the import side is a petroleum story with two rounding errors attached.

10. Trends and outliers compared

An outlier here is defined, not asserted. The workbook computes a z-score for each year of growth against that category's own mean growth, so a value above 2 marks a year that is unusual for that category rather than merely large.

Category	Outlier year	Growth that year	Z-score
8 Miscellaneous manufactured articles	2020	-10%	-2.41
6 Manufactured goods by material	2020	-16%	-2.37
7 Machinery and	2020	-14%	-2.15

transport equipment			
3 Mineral fuels	2022	+99%	2.15
9 Commodities not classified elsewhere	2025	+47%	2.52

The outliers fall into three groups and each has a different explanation. The 2020 group is the manufacturing categories, cut by the COVID demand collapse. The 2022 outlier is energy, lifted by the price shock after the invasion of Ukraine. The 2025 outlier is category 9, and that one is worth care. Category 9 rose $21,245m in 2025 while gold inside it rose $23,854m, so gold accounts for more than the whole category movement, offset by a $3,006m fall in confidential items. Gold rose 44% in price over 2025 against a 67% rise in Australian gold export value, so price explains most of the rise but not all of it. The remaining gap points to volume or refining throughput, which this dataset cannot separate.

11. Ten years of gain and loss, the target, and the forecast

The product chosen for this section is petroleum import, subcategory 33. It stood at $23,315m in 2016 against petroleum export of $6,946m, and by 2025 it had reached $52,742m against export of $14,187m. Across the ten years Australia lost ground on it. Import rose $29,427m, which is a rise of 126%, at a compound rate of 9.5% a year. The petroleum trade deficit widened by $22,186m over the same period. That is the loss, stated plainly: a larger bill every year for a product Australia increasingly does not refine for itself.

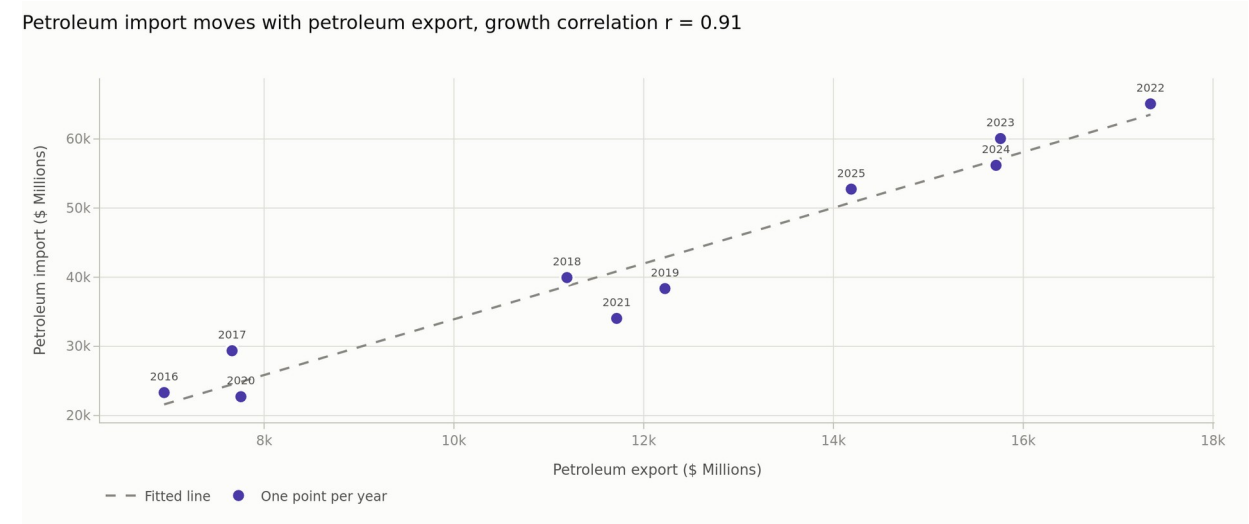


Figure 14. Petroleum import against petroleum export, one point per year, with the fitted line.

This chart carries the central finding. The two directions move together at r = 0.91. Both are priced off the same world oil market, so they rise and fall as one. Because import is about three times the size of export, a price rise adds roughly three dollars to the import bill for every dollar it adds to export earnings. Growing petroleum export cannot close this gap. Only buying less, or refining more at home, can.

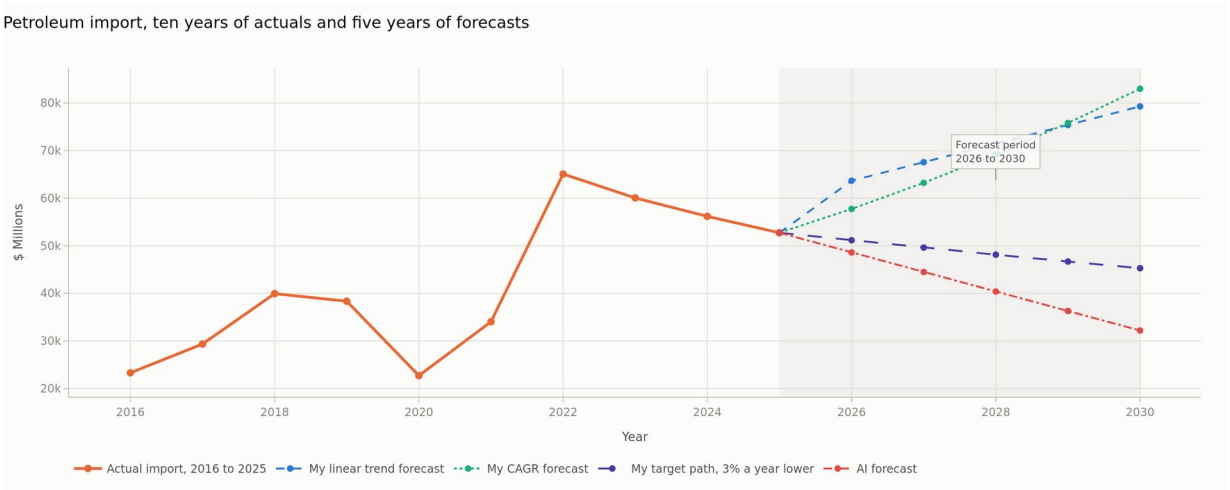


Figure 15. Petroleum import. Ten years of actual values, two forecast methods, the target path and the AI forecast, in $ Millions.

The workbook carries two forecast methods, both as live formulas. A straight line fitted to 2016 to 2025 gives $79,296m by 2030. Growing the 2025 value at the decade compound rate gives $83,007m. Both point upward and neither should be trusted far: the straight line explains only 59% of the variance, because this series follows a world price rather than a trend.

The target is therefore stated as a decision rather than a prediction. Cutting petroleum import 3% a year from the 2025 level reaches $45,291m in 2030, which is $34,004m below where the straight line would otherwise land.

Year	Linear trend	CAGR method	Target path	AI forecast
2026	63,668	57,749	51,160	48,600
2027	67,575	63,232	49,625	44,500
2028	71,482	69,236	48,136	40,400
2029	75,389	75,809	46,692	36,300
2030	79,296	83,007	45,291	32,200

12. Comparison against the AI forecast

An AI model was given the ten annual petroleum import values and the unit, and nothing else. It received none of this analysis, no change point notes, no refinery history, no export side and no target. The run used Claude Sonnet 5 at low effort on 16 October 2026, with no tools. The prompt and the reply are reproduced in full in the appendix.

Where the AI did well

1. It read the shape of the decade correctly, finding the 2022 peak, the 2020 dip and the three year decline after 2022.
2. It noticed that 2025 is still more than double the 2016 level.
3. It stated its method, extrapolating the 2022 to 2025 fall of about 4,113 million a year, so a reader can check the claim. It took a few seconds.

Where it went wrong

The AI fitted only the last four years. This analysis fits all ten. The two choices point in opposite directions. The AI forecasts $32,200m for 2030, a fall of 39% from 2025. The linear trend forecasts $79,296m, a rise of 50%. The gap between the two answers in 2030 is $47,096m, which is more than the entire 2016 import value.

What the AI could not know

- That the 2020 dip and the 2022 peak are price events rather than demand events. That only becomes visible from the two way data, where import and export move together at r = 0.91. The AI was given the import side alone.
- That two Australian refineries closed in 2021, which shifts the mix of imports from crude towards refined product and puts a floor under the bill.
- That the series is too noisy for a point forecast. The AI gave five exact numbers with no error range, which reads as far more certain than the data allows.

Advantages and disadvantages of using AI here

The advantages are real. The AI was fast, it was free of the anchoring that comes from having already built an opinion, and it stated a method that could be checked. Used as a second opinion it did something useful: it exposed that my own whole decade fit is itself an arbitrary choice of window.

The disadvantages are larger for this task. The AI had no causal knowledge and no way to acquire any, so it could not know why 2022 happened and therefore could not judge whether to extrapolate it. It gave confident point numbers with no uncertainty attached. It could not see the second trade direction, which is where the actual finding lives. The honest conclusion is that neither forecast deserves confidence, and that is the most useful thing the comparison produced.

13. Visualisation techniques, and the problems handled

The techniques were chosen per job: time series lines for trends, columns for a signed quantity such as the balance, small multiples where ten categories had to be compared at once, a dumbbell for a before and after comparison, a heatmap for the correlation matrix, and a scatter with a fitted line for the relationship between the two trade directions. Five problems came up and each needed a decision.

4. Ten categories, eight safe colours. Cycling hues past eight makes series indistinguishable, so the ten category charts became small multiples. Colour was then freed for the one distinction that matters everywhere, import against export.
5. Different magnitudes on one chart. Category 4 is about one hundredth the size of category 2. A second y axis was rejected, because two scales on one frame invite false comparison. Indexing to 2016 and small multiples solved it honestly.
6. Labels colliding. The first share chart put ten end of line labels on top of each other and was unreadable. It was rebuilt as a dumbbell on a category axis, with the values in a fixed column clear of the markers.
7. Grouped bars that did not group. The outlier chart first used a label of category and year, and because the outlier year differs by direction it produced twenty separate categories instead of ten pairs. The year moved to a bar label and the axis became the category alone.
8. Colour that fails for colourblind readers. The palette was not chosen by eye. Every set of colours that appears together was run through a contrast and colour vision checker, and two of the hues sit below the contrast floor on a light background, so every chart that uses them also ships a table of its numbers.

14. The dashboard against the storyboard

	Dashboard	Storyboard
What it is for	Looking something up	Being walked through an argument
Order of reading	Any order the reader likes	Fixed, top to bottom
Strength	Density and comparison on one screen, with filters	Causation and sequence, each panel sets up the next
Weakness	Shows what, not why. A reader can misread it with no guard rail	Slow to scan, and cannot answer a question it was not built to answer
Best audience	An analyst who already knows the domain	A policy reader meeting the data for the first time

The two methods failed in opposite ways while this task was built, which is the clearest evidence for using both. The dashboard shows that the petroleum deficit widened, but a reader can look at it and conclude that Australia should export more petroleum, which the data says will not work. The storyboard forces the scatter in section 11 in front of the reader before it allows any recommendation. The storyboard cannot be filtered, though, so a reader who wants 2020 to 2023 alone has to go to the dashboard. Together they cover both needs. Separately each leaves a real gap.

15. Recommendations

These recommendations cover the industries this study touches: refining, fuel import logistics, and the energy export industries that fund them.

9. Set the petroleum import bill as a tracked target, not a forecast. The path in this report cuts it 3% a year to $45,291m by 2030. A forecast on this series is not reliable enough to steer by, but a target is a decision and can be measured against each year.
10. Do not pursue petroleum export growth as a fuel security measure. Import and export move together at r = 0.91 and import is about three times larger, so a higher oil price widens the deficit no matter how much Australia sells. This is the single clearest policy finding in the data.
11. Treat the two remaining refineries as the variable that matters. Refining capacity is what decides whether Australia imports crude or finished product, and that mix is the part of the bill domestic policy can still influence.
12. Do not read the energy export surplus as durable cover for the fuel bill. Category 3 export has fallen three years in a row and has the highest volatility of any category, at a coefficient of variation of 0.38. The surplus that funded the import bill in 2022 has already shrunk 65%.
13. Diversify with care, because the two largest export categories do not hedge each other. Categories 2 and 3 correlate at r = 0.03, which looks like good diversification, but both are commodity price cycles. They were uncorrelated over this decade by timing, not by structure.
14. Collect volume alongside value. Every limit in this report comes from the dataset holding dollars only. Separating price from volume would settle several questions this analysis could only bound.

16. Conclusion

Ten years of Australian trade data show a surplus that peaked on a price shock and has fallen 65% since. Underneath it, subcategory 33 runs the other way. Australia buys far more petroleum than it sells, the gap has more than doubled, and the two sides of that trade move together closely enough that selling more cannot fix it. The analysis in this report is reproducible: every number lives in the submitted workbook as a formula, and the dashboard, the storyboard and this report all read that one file.

Appendix A. The AI prompt and the reply, word for word

Run on 16 October 2026 using Claude Sonnet 5, model id claude-sonnet-5, at low effort, with no tools available. The model was given the ten values below and no other information about this project.

The prompt

```
Australian imports of SITC subcategory 33, Petroleum, petroleum products and
related
materials. Annual values in millions of Australian dollars.

2016 23315
2017 29365
2018 39947
2019 38357
2020 22720
2021 34044
2022 65082
2023 60057
2024 56183
2025 52742

Do two things. Keep the whole answer under 200 words. Answer directly from the
numbers given. Do not use any tools.
```

```
1. Describe the trend between 2016 and 2025.
2. Forecast the value for each year from 2026 to 2030. Give one number per year in
millions of Australian dollars.
```

The reply

```
1. Imports rose overall from 2016 to 2025. Values climbed from 23315 in 2016 to a
peak
of 65082 in 2022, with a dip in 2020 to 22720, likely due to reduced demand that
year.
After 2022, values fell each year to 52742 in 2025, though 2025 is still more than
double the 2016 level.

2. Forecast for 2026 to 2030, based on the recent downward trend from 2022 to 2025
(an
average fall of about 4113 per year):

2026: 48600
2027: 44500
2028: 40400
2029: 36300
2030: 32200
```

Appendix B. Sources

Data: Australian Bureau of Statistics, Australian International Trade, 2016 to 2025, supplied as 2026-Spring-Ass3-V20.xlsx. External causes cited in section 6 and section 10 are recorded with their sources in data/research_notes.md, submitted with the dashboard source. The notes cover the 2020 COVID demand collapse, the 2022 energy price shock, the Kwinana and Altona refinery closures, China's coal restrictions, the iron ore price cycle and the 2025 gold price rise.